# Static and kinetic friction

**antocciphysics**

**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Distinguish adjustable static friction from sliding friction.

**SUCCESS CRITERIA - I can...**

1. Compare a push with the static-friction limit.
2. Find actual friction for rest or sliding.
3. Explain friction direction using relative slipping.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

## 01 Bring back an earlier idea

A spacecraft of mass 3 kg pushes an asteroid of mass 300 kg with 15 N. Compare interaction forces and the accelerations due to that interaction.

**Marzano check - circle:** ① ② ③

## 02 Explain a worked example

**Reference:** $0 \le f_s \le \mu_s N$; $f_k = \mu_k N$.
s = static; k = kinetic; N = normal force. Static friction adjusts up to its maximum. Kinetic friction applies during sliding.

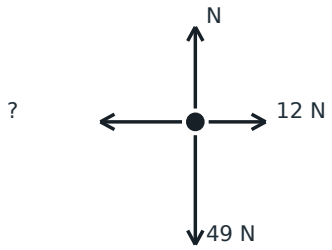
**Worked reasoning:** A box remains still while pushed with 3 N. Static friction is 3 N opposite the push, provided its maximum is at least 3 N. Static friction need not equal its maximum.

**Your turn:** Draw the horizontal forces. Then redraw for a 5 N push that is still below the maximum.

**Marzano check - circle:** ① ② ③

## 03 Actual versus maximum

A 5 kg box on a level floor has $\mu_s = .40$ and $\mu_k = .30$. Horizontal push 12 N; initially at rest. Find N, $f_s$, max, actual friction and a. Use g=9.8.




The dot represents the object. Initially at rest

**Marzano check - circle:** ① ② ③

## 04 Sliding case

Increase the push on that box to 25 N. Find friction and acceleration after it slides.

**Marzano check - circle:** ① ② ③

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### 05 Estimate from data

Supplied trials for normal force 20 N: peak forces 8, 9, 7 N; steady sliding forces 6, 5, 7 N. Find mean peak, mean sliding force, and coefficient estimates. Which coefficient describes threshold?

**Marzano check - circle:** 1 2 3

### 06 Choose a method

$x_i$=-4±0.5 m, v=6±0.1 m/s and exact t=8 s. Give central final position, its conservative interval and full width. Name an assumption.

**Marzano check - circle:** 1 2 3

### 07 Exit check

A 10 kg box has$\mu_s$=.50,$\mu_k$=.40, g=9.8. Compare 30 N and 60 N horizontal pushes, starting from rest each time. Give actual friction and acceleration.

**Marzano check - circle:** 1 2 3

### 08 STRETCH: Apply it another way

A 5 kg box has $\mu_s$=0.4 and $\mu_k$=0.3 on a level floor; use g=10 N/kg. A horizontal 22 N pull begins from rest. Find maximum static friction, acceleration after motion starts, and the pull needed for steady sliding.

**Marzano check - circle:** 1 2 3